

AVS-ESA: A Multi-Objective Optimization Approach for Energy-Efficient Variable-Speed Flexible Job Shop Scheduling

1st Shiliang Shao

School of automation and electrical engineeringg

Beihang University

Beijing, China

ssl020104@buaa.edu.cn

2nd Yunhong Xu

School of automation and electrical engineeringg

Beihang University

Beijing, China

susanbuaa1989@gmail.com

3rd Hongquan Qu

School of automation and electrical engineeringg

Beihang University

Beijing, China

qhongquan@buaa.edu.cn

4th Maolin Cai

School of automation and electrical engineeringg

Beihang University

Beijing, China

caimaolin@buaa.edu.cn

5th Yong Hu

Shengu Group Co. LTD

Shenyang, China

huyong@shengu.com.cn

6th Xiaomeng Tong

School of automation and electrical engineeringg

Beihang University

Beijing, China

tongxiaomeng@buaa.edu.cn

Abstract—As the demand for industrial products surges rapidly, industry accounts for over 45% of global energy consumption. Therefore, an effective scheduling plan can contribute to reducing carbon emissions. However, current research on the energy-aware Flexible Job Shop Scheduling Problem (EFJSP) under variable speeds primarily focuses on energy consumption at the job level, neglecting the energy consumption of operations under different constraints—a factor with critical environmental implications. In this paper, we investigate the scheduling problem under variable speed conditions, with the primary objective of reducing manufacturing energy consumption without altering production time. To address this problem, we introduce processing quality as a constraint to define the upper limit of variable speeds for the first time and design a simplified variable-speed flexible job shop scheduling environment system. To evaluate this system, we propose the Adaptive Variable-Speed Energy-Saving Scheduling Algorithm (AVS-ESA), a novel model and algorithmic framework to solve the problem and compare it with the exact solver OR-Tool and CMOEA. The results demonstrate that our proposed algorithmic framework effectively reduces energy consumption without compromising manufacturing time.

Index Terms—energy-aware flexible job shop scheduling, multi-objective optimization, variable speed, processing quality.

I. INTRODUCTION

Compared to the traditional Job Shop Scheduling Problem (JSSP), the Flexible Job Shop Scheduling Problem (FJSP) and its extension, the EFJSP, represent a more complex NP-hard problem. This complexity arises from the assignment of jobs to different processing machines. EFJSP introduces additional constraints related to machine-specific limitations

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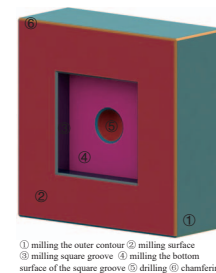


Fig. 1. Example of multiple operations in a job

for different jobs, further complicating multi-objective optimization. In prior studies on JSSP, the relationship between jobs and machines is typically decomposed into two independent stages. The first stage, known as assignment, involves allocating operations to one of several available machines. The second stage, operation sequencing, entails adjusting the order of operations on the same or different machines to achieve an optimal overall solution. Existing research on the variable-speed flexible job shop scheduling problem aims to optimize energy consumption as one of the multiple objectives, often assuming uniform energy consumption across operations within the same job. However, in a variable-speed context, the energy consumption of operations is closely tied to processing speed. Specifically, variations in processing speed can impact total energy consumption (TEC), processing quality, and even the makespan (MS). As illustrated in Fig. 1, completing a job requires multiple operations, each exhibiting distinct energy consumption levels under different speeds.

To address the variable-speed Energy-aware Flexible Job Shop Scheduling Problem (EFJSP), we propose the Adaptive Variable-Speed Energy-Saving Scheduling Algorithm (AVS-ESA). This novel model integrates the selection of processing speeds, requirements for processing quality, and heuristic energy-saving scheduling strategies. AVS-ESA facilitates practical energy-efficient scheduling solutions by choosing appropriate processing speeds to meet quality requirements without increasing the Makespan (MS) while simultaneously reducing Total Energy Consumption (TEC). In the subsequent sections, we will introduce our multi-objective optimization model. By considering the objectives of minimizing both MS and TEC, we can derive more practical, efficient, and energy-saving scheduling solutions.

II. LITERATURE REVIEW

According to statistics from the International Energy Agency, the industrial sector accounted for approximately 46% of global final energy consumption in 2021 [1]. Consequently, research on the EFJSP has garnered increasing attention. Building on the work of other researchers, we summarize classical studies aimed at reducing TEC and highlight recent advancements in this field. Brandimarte and Calderini (1995) proposed the integration of process planning and shop scheduling, laying a theoretical foundation for subsequent research combining energy consumption optimization with shop scheduling [2]. Yan et al. (2016) developed an optimization approach based on job-specific energy consumption, establishing energy models for both machine tools and the workshop. By enhancing the NSGA-II algorithm, they optimized the processing procedure, yielding reasonable processing and scheduling solutions [3]. Dai et al. (2019) treated energy consumption as one of the optimization objectives, incorporating job transportation time. Using a multi-objective genetic algorithm, they significantly reduced TEC while meeting production demands through optimized scheduling [4]. An et al. (2020) examined the impact of varying processing resources on energy consumption, employing a decomposition-based evolutionary algorithm to solve for optimal MS and TEC [5]. Öztö et al. (2020) proposed a bi-objective mixed-integer programming model in a variable-speed processing environment and designed a multi-objective algorithm based on the epsilon-constraint method to obtain Pareto-optimal solutions [6]. Abedi et al. (2020) considered machine deterioration rates and processing time variations due to variable speeds, proposing a population-guided multi-objective optimization algorithm that effectively reduced MS and TEC in this scenario [7]. Wang and Wang (2022) developed a memetic algorithm based on learning agents, demonstrating superior performance in large-scale instances [8]. Qu et al. (2024) introduced an improved decomposition-based multi-objective optimization algorithm under variable processing speeds, effectively reducing TEC without compromising MS [9]. Fontes et al. (2024) addressed a scenario integrating transportation time and variable processing speeds, developing a stochastic critical genetic algorithm that efficiently identified

high-quality solutions [10]. Ham et al. (2021) designed a mixed-integer programming model combined with constraint programming, flexibly incorporating turn off strategies, with results outperforming other models [11]. Akram et al. (2024) tackled dynamic EFJSP by designing a framework with flexible turn offs, proposing a black widow spider algorithm that generated excellent Pareto fronts in data analysis [12]. The aforementioned studies address EFJSP across diverse scenarios, employing a variety of methods. From considering processing techniques and transportation times to designing models for dynamic environments, these approaches utilize variable-speed and machine on/off strategies. They offer distinct perspectives and innovative techniques for optimizing MS and TEC.

III. PROBLEM PRESENTATION

In JSSP, all machines are typically available for all jobs; however, in FJSP, certain operations are unavailable on specific machines. For the EFJSP with variable processing speeds, existing studies often lack clear definitions for the upper and lower bounds of speed variation. In real-world production, variable speeds are closely tied to processing quality, where higher speeds may reduce time but often compromise quality, which could be unacceptable to buyers. To address this, it is necessary to define quality requirements and machine load specifications for jobs in variable-speed EFJSP, thereby establishing appropriate processing speed bounds and enabling the selection of suitable operations for variable-speed execution. Our objective is to identify an optimal scheduling solution among available machines that satisfies these quality requirements, requiring a precise definition of the problem's fundamental constraints and parameters

A. Notation Definition

To facilitate the description of the model design, we use symbols to represent certain components. The notation is defined as follows:

- n : total count of jobs.
- i : identifier for jobs, $i = 1, 2, \dots, n$
- n_i : quantity of operations within each job.
- j : sequence number of operations for each job, $j = 1, 2, \dots, n_i$.
- $O_{i,j}$: the j -th task in the i -th job.
- m : total number of machines available.
- k : machine identifier, $k = 1, 2, \dots, m$.
- M_k : designation for M_k .
- $V_{i,j,k}$: maximum speed that ensures the processing quality of $O_{i,j}$ on M_k .
- s : indicator of the processing speed for each task, $s = 0.5, 0.6, \dots, V_{i,j,k}$.
- $s_{i,j,k}$: a continuous variable representing the selected speed multiplier for operation $O_{i,j}$ on machine M_k .
- $X_{i,j,k}$: A binary variable; $X_{i,j,k} = 1$ if operation $O_{i,j}$ is assigned to machine M_k , and 0 otherwise.
- $e_{i,j,k}$: coefficient of energy usage acceleration for $O_{i,j}$ on M_k .
- u_i : marker for idle periods on a machine.
- p : marker for active processing periods on a machine.
- L_k : workload assigned to M_k .

J_n : identifier for job n .
 $Q_{i,j}$: required quality standard for task $O_{i,j}$.
 $C_{i,j}$: time at which task $O_{i,j}$ is completed.
 $T_{i,j,k}$: duration required to process $O_{i,j}$ at the standard speed on M_k .
 $TI_{u,k}$: the u -th idle duration on M_k .
 $TM_{k,p}$: the p -th active processing duration on M_k .
 MP_k : process power on M_k .
 TMS_k : safe time gap on M_k .
 MSE_k : energy required for starting up M_k .
 MIP_k : power consumption during idle state on M_k .
 MWE_k : energy used during the warm-up phase on M_k .
 $MSTE_k$: energy needed for setup on M_k .
 $MSBE_k$: energy consumption in standby mode on M_k .
 $PO_{i,j,k,s}$: actual energy consumed per unit for $O_{i,j}$ at speed s on M_k .
 $N_{\text{setup},k}$: total number of startup events for machine M_k during the scheduling horizon.
 EMA_k : aggregate energy cost for machine startup on M_k .
 TEC : total energy consumption.
 E_{process} : total energy consumed during the active processing of all operations.
 E_{idle} : total energy consumed by all machines during idle periods.
 E_{setup} : total energy consumed by all machine startup events based on the on/off strategy.
 C_{max} : the makespan, i.e., the completion time of the last operation.

B. Model Definition

Based on the symbols defined above and the framework developed by Qu et al [9]. The EFJSP under variable speed conditions, as discussed in this paper, can be mathematically formulated as follows. This model aims to achieve dual-objective minimization of makespan (C_{max}) and TEC by collaboratively optimizing the scheduling scheme and processing speeds.

The objective functions are defined as:

$$\min f_1 = C_{\text{max}} \quad (1)$$

$$\min f_2 = TEC \quad (2)$$

The TEC is composed of three primary components: E_{process} , E_{idle} , and E_{setup} . These components are defined as follows:

$$TEC = E_{\text{process}} + E_{\text{idle}} + E_{\text{setup}} \quad (3)$$

$$E_{\text{process}} = \sum_{i=1}^n \sum_{j=1}^{n_i} \sum_{k=1}^m X_{i,j,k} \cdot PO_{i,j,k,s} \cdot \frac{T_{i,j,k}}{s_{i,j,k}} \quad (4)$$

$$E_{\text{idle}} = \sum_{k=1}^m \sum_u MIP_k \cdot TI_{u,k} \quad (5)$$

$$E_{\text{setup}} = \sum_{k=1}^m N_{\text{setup},k} \cdot EMA_k \quad (6)$$

Compared to existing EFJSP models in the literature, the core distinction and innovation of this model lies in its

pioneering use of processing quality as a primary constraint to directly define the upper limit of variable processing speeds. Most extant models, when addressing variable-speed energy saving, either do not explicitly define the boundaries of speed variation or treat it merely as a general parameter. This model tightly couples the abstract quality requirement $Q_{i,j}$ with the concrete process parameter $s_{i,j,k}$ through the constraint $s_{\min,i,j} \leq s_{i,j,k} \leq V_{i,j,k} \quad \forall i, j, k$, enhancing the practical feasibility of the optimization results in real-world industrial scenarios. Furthermore, this model not only considers the nonlinear impact of speed variation on energy consumption (reflected in $PO_{i,j,k,s}$) but also introduces a judgment mechanism based on machine average load to determine the application of the variable-speed strategy. This approach prevents overloading individual machines in pursuit of localized energy savings, thereby making the resulting scheduling solutions more balanced and robust.

C. FJSP System Environment Design

EFJSP is an extension of FJSP, necessitating the initial design of a flexible job shop scheduling system environment as the foundation for the EFJSP environment. We designed a system, as shown in Fig. 2, comprising 5 machines and 5 jobs, with each job consisting of different operations. To reflect the characteristics of a flexible job shop, each operation can be assigned to different machines for processing; however, not all machines are permitted to process the same operation.

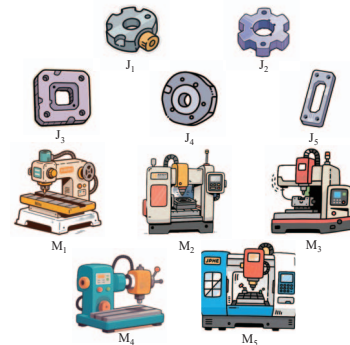


Fig. 2. Designed FJSP system environment

In this system environment, each job is composed of several operations, with strict precedence constraints between operations. Specifically, a subsequent operation within a job can only begin after its preceding operation has been completed. Meanwhile, different jobs are independent of one another, with no dependencies between their respective operations. Each machine can process only one operation at a time during the processing phase, and interruptions are not allowed—once an operation begins, it must be completed continuously. Additionally, the system assumes that all jobs and machines are available at the start of scheduling, with no initial delays or failures.

By predefining the parameters shown in Table I within the FJSP environment, we establish a foundation for extending

the system to EFJSP. In Table I, the processing times marked in bold represent the time required for the selected machine to process the current operation. This design ensures the core characteristic of assignment in the FJSP, and meanwhile provides data support for multi - objective optimization in the EFJSP.

TABLE I
PROCESSING TIMES OF JOBS ON DIFFERENT MACHINES

Job	Operation	Machine				
		M_1	M_2	M_3	M_4	M_5
J1	$O_{1,1}$	-	4	5	-	5
	$O_{1,2}$	5	5	-	-	6
	$O_{1,3}$	2	3	3	6	-
J2	$O_{2,1}$	5	4	-	-	-
	$O_{2,2}$	-	3	-	3	5
J3	$O_{3,1}$	4	2	3	-	-
	$O_{3,2}$	6	-	5	9	-
	$O_{3,3}$	-	2	-	-	3
J4	$O_{4,1}$	5	-	-	4	5
	$O_{4,2}$	-	4	3	-	-
	$O_{4,3}$	5	-	3	-	4
J5	$O_{5,1}$	3	5	5	-	6
	$O_{5,2}$	4	-	5	-	6
	$O_{5,3}$	4	-	-	5	5
	$O_{5,4}$	2	-	-	-	3

IV. EFJSP MODELING

Building on the system environment of FJSP, we establish an energy-aware model, requiring the definition of additional parameters to achieve the goal of identifying the most reasonable scheduling solution. These parameters encompass factors such as job quality, processing speed limits, and machine processing strategies, forming the basis for solving the optimal scheduling solution space.

TABLE II
QUALITY REQUIREMENTS OF JOBS AND CORRESPONDING PARAMETERS

Job	Operation	Parameters					
		Q	s_{max}	s_{min}	PO	PO_{smax}	PO_{smin}
J1	$O_{1,1}$	1	1	0.5	5	5	2
	$O_{1,2}$	7	1.6	0.7	8	10	6
	$O_{1,3}$	1	1	0.5	7	7	2
J2	$O_{2,1}$	11	2	1	5	9	5
	$O_{2,2}$	8	1.7	0.8	5	11	4
J3	$O_{3,1}$	7	1.6	0.5	6	8	3
	$O_{3,2}$	10	1.9	0.6	6	8	2
	$O_{3,3}$	1	1	1	3	3	3
J4	$O_{4,1}$	1	1	1	5	5	5
	$O_{4,2}$	1	1	0.5	10	10	8
	$O_{4,3}$	4	1.3	1	7	9	7
J5	$O_{5,1}$	3	1.2	1	3	4	3
	$O_{5,2}$	1	1.1	0.6	9	12	7
	$O_{5,3}$	7	1.6	0.5	5	6	1
	$O_{5,4}$	11	2	0.9	6	12	4

A. Job Information

The quality requirements of the system environment, processing speed, and corresponding energy consumption are

defined in Table II. The processing quality imposes an upper limit on the processing speed, which must satisfy the following constraint:

$$s_{actual} \leq f(Q) \cdot s \quad (7)$$

where $f(Q)$ is determined by the process specification sheet of the actual product type being processed.

In the designed scheduling system, we represent the processing speed as a multiple of the default processing time. For example, for operation $O_{5,4}$, the s_{max} indicates that the processing speed can reach up to twice the original speed, reducing the corresponding processing time from 2 (as shown in Table I) to 1. Processing speed directly influences energy consumption: faster speeds result in shorter processing times but increase energy consumption. Notably, the variation in processing energy consumption is typically nonlinear.

B. Machine Information

TABLE III
MACHINE PARAMETERS

Machine	Parameters					
	MSE_k	MWE_k	$MSBE_k$	$MSTE_k$	MIP_k	TMS_k
M_1	2	1	2	3	5	3
M_2	2	5	5	6	7	3
M_3	5	2	3	2	6	5
M_4	3	4	5	5	8	4
M_5	4	3	2	3	3	3

The machine parameter settings within the system environment are defined in Table III, which directly impact the application of energy-saving strategies. To simplify the description, we define $EMA_k = MSE_k + MWE_k + MSTE_k + MSBE_k$.

V. ALGORITHM SOLUTION AND RESULT

A. Definition of Energy-Saving Strategies

To ensure machines achieve an optimal energy consumption scheme while processing operations, energy-saving strategies must be defined. A critical component is the machine on/off strategy. In Table III, we define TMS_k for each machine, indicating that if the time window between operations exceeds TMS_k , turning off the machine can be considered. When processing operations at variable speeds, prioritizing the best energy consumption scheme while ignoring machine load is impractical. Therefore, by calculating the average machine load, we constrain whether a machine can process an operation at variable speed. The constraint is expressed as follows:

$$T_{i,j,k} = \begin{cases} \frac{1}{s} T_{i,j,k}, & \text{if } L_k - \frac{\sum^m L_k}{m} < 0 \end{cases} \quad (8)$$

Here, if the judgment condition is less than 0, it indicates that the load of machine M_k does not exceed the average load across all machines, allowing variable-speed processing. If the condition is greater than 0, variable-speed processing is not permitted. Examples for explaining the machine on/off strategy and the variable-speed strategy are shown in Fig. 3 and Fig. 4 respectively.

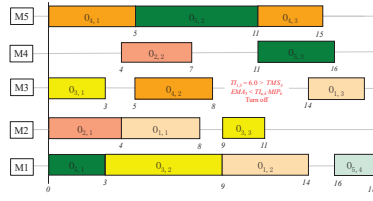


Fig. 3. Explanation of the machine on/off strategy

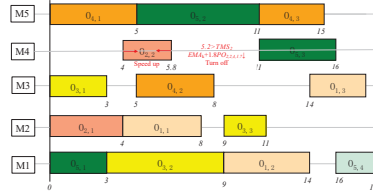


Fig. 4. Explanation of the variable-speed strategy

B. Algorithm Steps

Several existing multi-objective optimization algorithms can address EFJSP. We apply the aforementioned operators to the variable-speed EFJSP, using the classic NSGA-II as the foundational framework. A neighborhood structure is employed to optimize the scheduling scheme, ultimately minimizing TEC through these operators. The detailed algorithm is described as follows:

- Step 1: Encoding—initialize the solution population, perform crossover and mutation, and decode.
- Step 2: Adjust the neighborhood structure and decode.
- Step 3: Label the quality $Q_{i,j}$ of all individuals.
- Step 4: Traverse all individuals; if $TI_{u,k} \geq TMS_k$ and $MIP_k \cdot TI_{u,k} > EMA_k$ (condition 1) are satisfied, the turn off strategy can be executed.
- Step 5: If condition 1 is not met, evaluate whether both ends of $Q_{i,j}$ can be accelerated to satisfy condition 1 based on $Q_{i,j}$. If acceleration satisfies the condition, accelerate $Q_{i,j}$ and mark the adjusted $Q_{i,j}$.
- Step 6: Traverse all individuals; if M_k satisfies the judgment condition of (8) and has not been marked as accelerated, perform a deceleration operation, ensuring the adjusted $Q_{i,j}$ still satisfies $Q_{i,j}$.
- Step 7: Update L_k .
- Step 8: If L_k changes, return to Step 6.
- Step 9: Based on fast non-dominated sorting, obtain the highest-level non-dominated population PF .

C. Comparison of Scheduling Results

For small-scale scheduling systems, exact solvers can compute the optimal scheduling solution. However, for medium-to large-scale scheduling systems, particularly those involving multi-objective optimization, exact solvers struggle to produce precise solutions within a reasonable timeframe. Given the small-scale scheduling system designed earlier, the inclusion of variable-speed operations makes it challenging for an exact solver to compute directly. Thus, we treat it as an FJSP

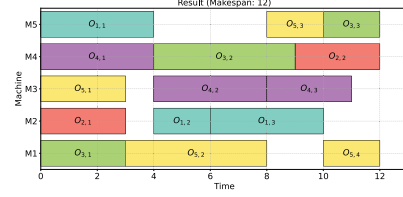


Fig. 5. MS calculated by OR-Tools

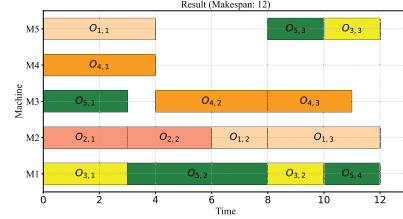


Fig. 6. MS calculated by the proposed algorithm

problem and solve it using an exact solver. We then solve the extended EFJSP problem for this scheduling system using our proposed algorithm and compare the results. As shown in Fig. 5 and Fig. 6, the MS for both approaches equals 12, indicating that, for this small-scale scheduling system, the MS results from our proposed algorithm align with those of the exact solver.

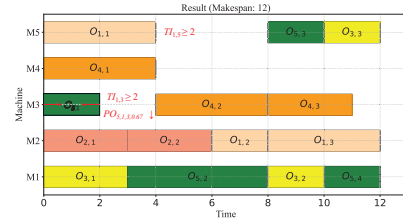


Fig. 7. TEC calculated by the proposed algorithm

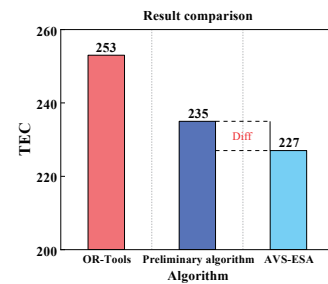


Fig. 8. Comparison of three results

To extend this to the EFJSP problem, we introduce the following assumptions for energy-related components: $TMS_k = 2$, $MP_k = 5$ kW, $MIP_k = 1$ kW, and $EMA_k = 1$ kWh. We use a fixed $s_{i,j,k} = 0.67$ and apply only energy-saving strategies for comparison. Based on these assumptions, the TEC for Fig. 5 is calculated as $TEC = E_{process} + E_{idle} = 245 + 8 = 253$ kWh. The TEC for Fig. 6 is $TEC = E_{process} + E_{idle} = 230 + 5 = 235$ kWh.

TABLE IV
COMPARISON RESULTS OF STANDARD DATASETS

Intances	OR-Tools		CMOEA		AVS-ESA	
	Makespan	TEC	Makespan	TEC	Makespan	TEC
MK01	40	939.8	42.1	867.4	40	854.2
MK02	28.2	770.3	34.2	766.9	28.1	745.7
MK03	204	6665.3	213.5	7125.6	226.5	6582.8
MK04	63.2	2074.4	62.4	1876.3	64.2	1865.9
MK05	174	3435.5	181.2	3452.2	173	3444.4
MK06	61.6	2371.9	67.3	4021.6	61.4	2003.6
MK07	142	3466.3	166.8	2898	140.4	2971.8
MK08	525	17864.2	533	19481.9	523	17046.5
MK09	309.1	17502.5	311.7	18319.5	308.2	16166.2
MK10	213.9	17558	218.6	19199.8	215.7	18431.5
lar03_1	278.1	29071.5	281.5	30537.2	277.3	25478.7
lar03_2	274.1	28552.9	277	33555.5	272.7	27059.8
lar03_3	282.8	32944.7	281.9	30734.7	269.8	27010.2
lar04_1	781.3	76869.6	643.4	87381.5	514.9	68342.5
lar04_2	807.7	67755.7	667.2	76767.2	501.8	59060.6

kWh. By further optimizing the result of Fig. 6 using our proposed operators, the outcome is shown in Fig. 7. Here, the idle time between $O_{1,1}$ and $O_{5,3}$ triggers the turn-off strategy based on $TI_{1,5} \geq 2$. $O_{5,1}$ is accelerated according to quality requirements. Other idle times are also subject to the turn-off strategy as appropriate. This yields a TEC of $E_{process} + E_{idle} + E_{setup} = 215 + 10 + 2 = 227$ kWh. The final TEC comparison is presented in Fig. 8. Our preliminary algorithm reduces energy consumption by 18 kWh compared to OR-Tools. Further optimization with our operators reduces energy consumption by another 8 kWh. This indicates that our proposed algorithm not only achieves the same makespan as exact solvers while maintaining production time, but also demonstrates superior performance in energy saving, offering significant guidance for energy management in real-world industrial production.

In addition to the comparative experiments conducted on the small-scale scheduling system described above, we further validated our approach using modified Brandimarte [13] and Behnke [14] benchmark datasets. These modified datasets simulate the EFJSP scenarios presented in this paper. Power consumption parameters were randomly assigned as follows: $PO_{i,j,k,s} \in (2,3)$, $EMA_k \in (1,2)$, and $E_{idle} \in (1,2)$. We compared our algorithm not only with the exact solver OR-Tools but also with the state-of-the-art multi-objective optimization algorithm CMOEA [15]. For each algorithm, the experimental parameters were set identically, and each experiment was repeated ten times, with the average value taken as the final result. The experimental results are presented in Table IV. The results demonstrate that in terms of makespan, the AVS-ESA algorithm achieved comparable or even superior performance to OR-Tools across most test instances, indicating AVS-ESA's competitiveness in maintaining or reducing production time. More importantly, regarding energy consumption, the AVS-ESA algorithm significantly outperformed both OR-Tools and CMOEA in nearly all test instances. This robustly proves the superior performance of the proposed AVS-ESA algorithm in minimizing energy consumption, particularly under the premise of guaranteeing or optimizing makespan, thereby

substantially reducing the total energy consumption during the manufacturing process. These findings further validate the substantial potential of the AVS-ESA algorithm for enhancing energy efficiency in real-world industrial production.

VI. CONCLUSIONS

To minimize TEC while maintaining the MS, the AVS-ESA algorithm imposes rational constraints on the variable speed range by incorporating processing quality and determining acceleration conditions by preserving machine load. Experimental simulations and an improved benchmark dataset validate the algorithm's feasibility and superior performance. Ultimately, the results confirm that AVS-ESA enhances energy efficiency without compromising manufacturing time.

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